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Domain :- AI - ML

Branch Name :- Computer Engineering

1)

```
df.head()
```

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
0	female	group B	bachelor's degree	standard	none	72	72	74
1	female	group C	some college	standard	completed	69	90	88
2	female	group B	master's degree	standard	none	90	95	93
3	male	group A	associate's degree	free/reduced	none	47	57	44
4	male	group C	some college	standard	none	76	78	75

Next steps: [New interactive sheet](#)

2)

```
df.shape
```

(1000, 8)

3)

```
df.columns
```

Index(['gender', 'race/ethnicity', 'parental level of education', 'lunch', 'test preparation course', 'math score', 'reading score', 'writing score'], dtype='object')
--

 df.dtypes

***	0
gender	object
race/ethnicity	object
parental level of education	object
lunch	object
test preparation course	object
math score	int64
reading score	int64
writing score	int64

dtype: object

5)

df.isnull().sum()

	0
gender	0
race/ethnicity	0
parental level of education	0
lunch	0
test preparation course	0
math score	0
reading score	0
writing score	0

dtype: int64

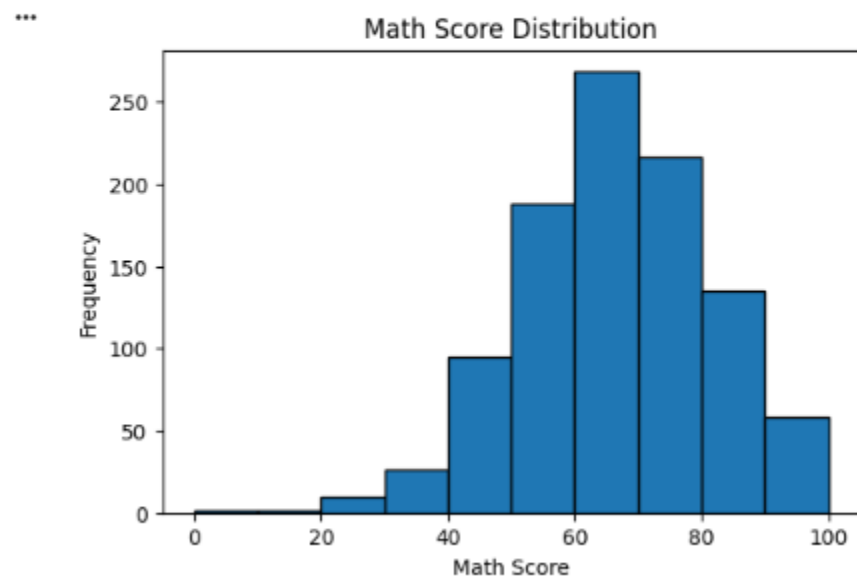
6)

```
df.describe()
```

	math score	reading score	writing score
count	1000.00000	1000.000000	1000.000000
mean	66.08900	69.169000	68.054000
std	15.16308	14.600192	15.195657
min	0.00000	17.000000	10.000000
25%	57.00000	59.000000	57.750000
50%	66.00000	70.000000	69.000000
75%	77.00000	79.000000	79.000000
max	100.00000	100.000000	100.000000

7)

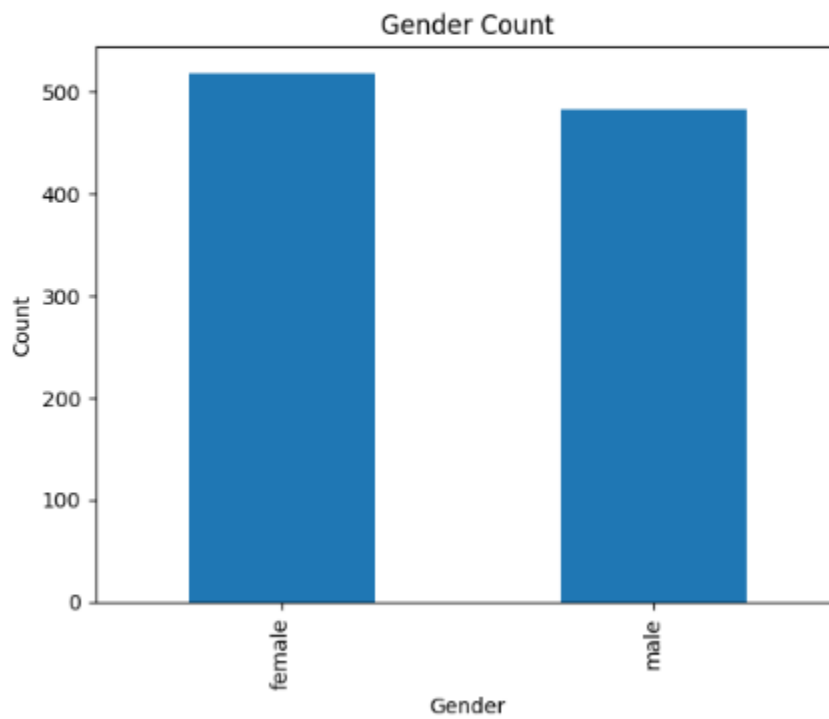
```
plt.figure(figsize=(6,4))
plt.hist(df["math score"], bins=10, edgecolor="black")
plt.title("Math Score Distribution")
plt.xlabel("Math Score")
plt.ylabel("Frequency")
plt.show()
```



8)

```
df["gender"].value_counts().plot(kind="bar")  
plt.title("Gender Count")  
plt.xlabel("Gender")  
plt.ylabel("Count")  
plt.show()
```

...



9)

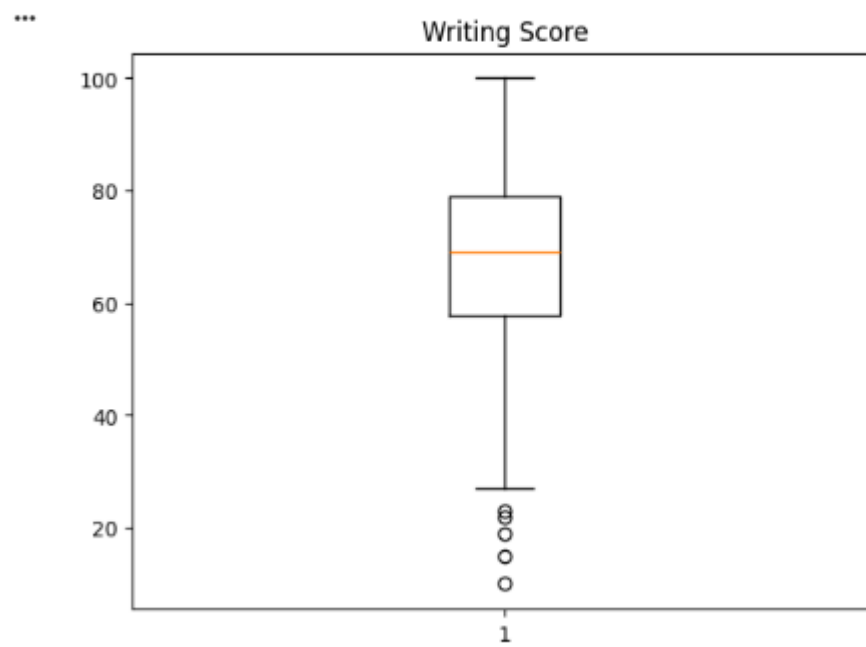
```
plt.scatter(df["math score"], df["reading score"])  
plt.title("Math vs Reading Score")  
plt.xlabel("Math Score")  
plt.ylabel("Reading Score")  
plt.show()
```

...



10)

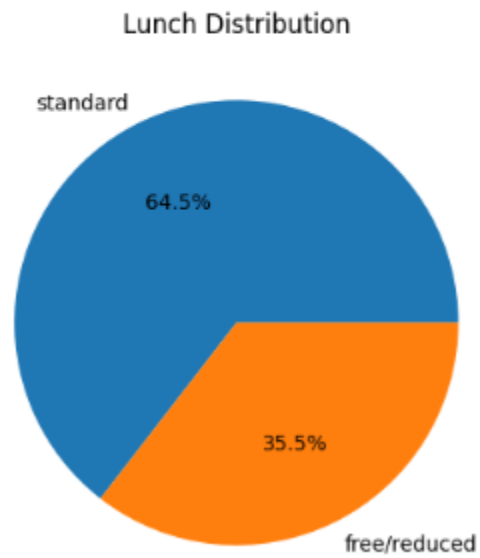
```
plt.boxplot(df["writing score"])  
plt.title("Writing Score")  
plt.show()
```



11)

```
df["lunch"].value_counts().plot(kind="pie", autopct="%1.1f%%")  
plt.title("Lunch Distribution")  
plt.ylabel("")  
plt.show()
```

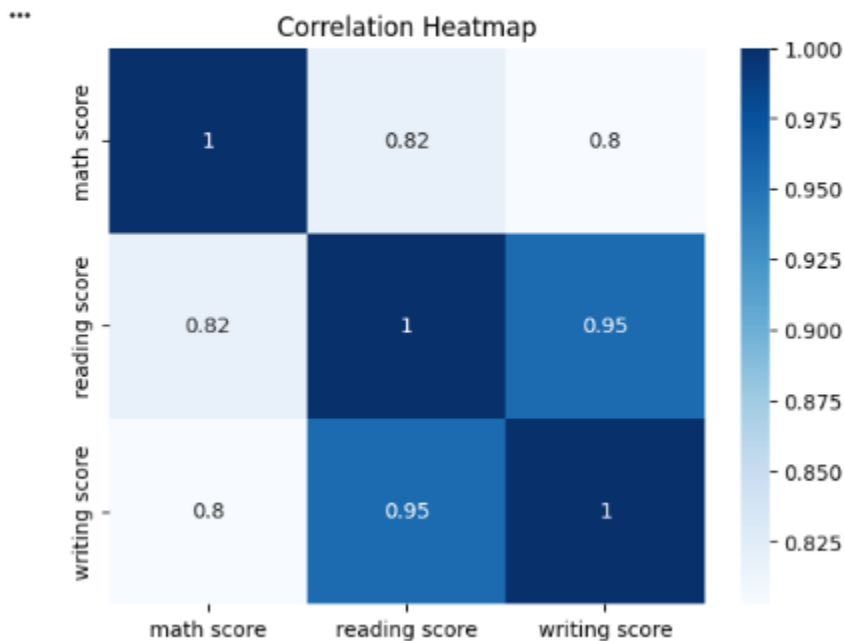
...



12)

```
import seaborn as sns

sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="Blues")
plt.title("Correlation Heatmap")
plt.show()
```



Brief Summary of dataset

- Dataset contains **1000 rows** and **8 columns**.
- No missing values are present.
- Students' scores are available in **Math, Reading, and Writing**.
- Male and Female students are almost equally represented.
- Math, Reading, and Writing scores are positively correlated

Dataset name

Student Performance

Number of rows and columns

Rows : 1000

Columns :- 8

Conclusion

The **Students Performance Dataset** was successfully analyzed using Python and Pandas. The dataset is clean, with no missing values, and the visualizations provided useful insights into student performance.